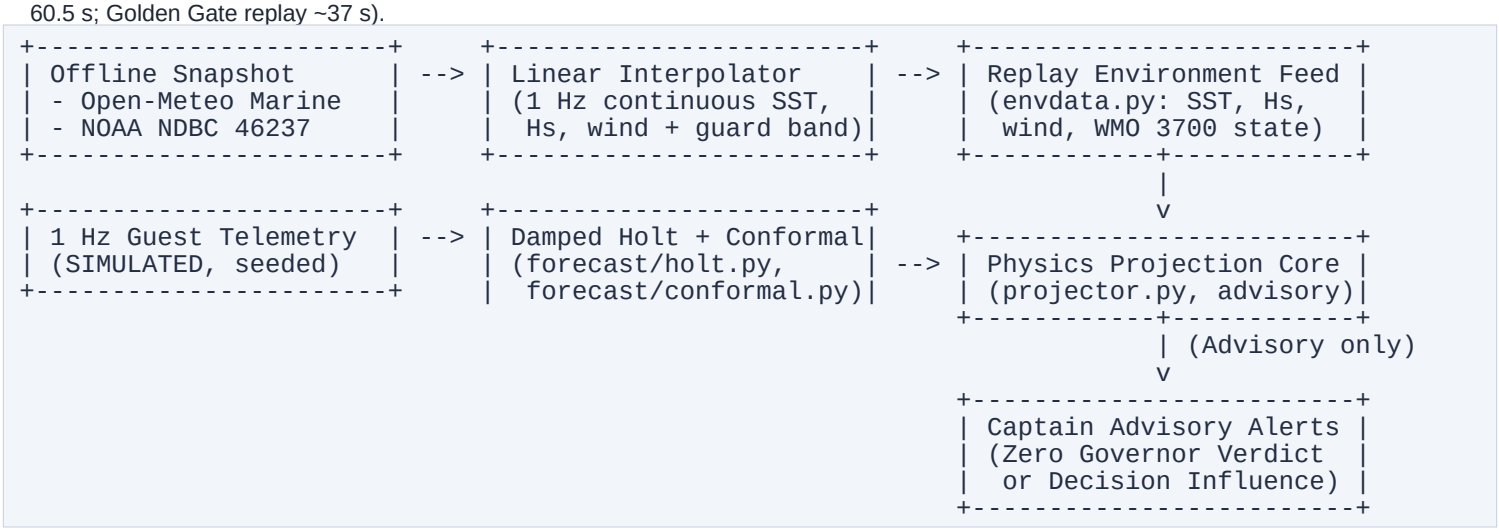
Technical Report: Public Metocean Data & Advisory Consumption Forecasting

This technical note documents the mathematical formulation, operational architecture, verification gates, and honest operational boundaries of the public environmental data ingestion and advisory load forecasting subsystem (sdv-resource-governor).

1. Architecture & Non-Interference Boundary

The forecasting and public environmental subsystem consists of two decoupled functional blocks designed for offline, deterministic naval governor execution:

* 1. Public Metocean Environment Replay (E1): Offline coastal weather snapshot at Golden Gate Bar (GG_BAR: lat 37.788, lon -122.634) combining Open-Meteo marine models with NOAA NDBC buoy ground truth (46237). Injects recorded sea surface temperature (SST), wave height (Hs), sea state (WMO Code Table 3700), and sustained wind into replays (envdata.py), and computes guarded, fail-closed operational windows over the 48 h outlook (env_windows.py); missing wind or sea state closes the window.
* 2. Predictive Consumption Advisory Observer (E2): Recursive 1 Hz time-series forecaster with conformal uncertainty intervals and exact physical budget projections (forecast/). Issues advisory warnings ahead of reactive shedding (benchmark median lead 60.5 s; Golden Gate replay ~37 s).



> Mandatory Non-Interference Principle (FC-INT-01..03): The observer is instantiated with observer=None by default and executed strictly outside the governor's authoritative state transitions and decision logic. Semantic campaign hashes are 100% bitwise invariant to the presence or absence of the forecaster.

2. Formulation

2.1 Normalized Damped Holt Model (forecast/holt.py)

Power is normalized by the declared mode rating, z = y / declared_w, so a 1 kW sensor and a 45 kW container are comparable. With damping phi < 1:

* level: l(t) = alpha z(t) + (1 - alpha)(l(t-1) + phi b(t-1))
* trend: b(t) = beta (l(t) - l(t-1)) + (1 - beta) phi b(t-1)
* forecast: z(t+h) = l(t) + b(t) (phi + phi^2 + ... + phi^h), for h in {10, 20, 30, 45, 60} s; y(t+h) = declared_w x z(t+h)

States: 20-sample WARMUP (ASM-FCST-WARMUP); trend reset on a mode change (ASM-FCST-MODE-RESET); gap > 3 s -> STALE, band x1.5 (ASM-FCST-STALE); gap > 10 s -> UNAVAILABLE (ASM-FCST-UNAVAIL); 3 consecutive residuals > 4 sigma -> REGIME_BREAK (ASM-FCST-REGIME).

2.2 Split-Conformal Bands (forecast/conformal.py)

For each horizon h the residuals z(t+h) - forecast are tabulated on the calib split; finite-sample corrected 5th and 95th percentiles q_lo(h), q_hi(h) give the 90 % band [y(t+h) + declared_w x q_lo(h), y(t+h) + declared_w x q_hi(h)].

2.3 Physical Budget Projection (forecast/projector.py)

* 1. Scalar pre-filter (every tick): is the upper bound > 1.10 x declared_w (ASM-FCST-OVERRUN-THRESHOLD)? If not, cost < 0.1 ms.
* 2. Bisection on the real budget engine: a cloned state is run through the eight-class engine (budgets.py); bisection over h in (0, 60] s on the upper trajectory finds the first infeasible horizon in at most 8 evaluations; point/lower crossings are a scalar scan against that level. If the point forecast never reaches it, the output says so (point_forecast_crosses = false).
* 3. Thermal inertia: for cooling witnesses, steps the lumped reservoir (1.8 MJ, 30 kW x 60 s) to the second it is exhausted.

3. Pre-Registered Acceptance Gates & Empirical Results

Measured by tools/bench_forecast.py --split both on the production ForecastObserver (on_sample + on_tick), commit c6c1052, macOS arm64. Disjoint seeds: train 1000--1239, calib 2000--2079, test 3000--3119, ood 4000--4119, control_steady_1h 5000--5005. Full output: docs/evidence/forecast_report_20261018.{md,json}. Re-run on the Arduino Ventuno Q (aarch64, same code_sha256 99ef1677? as Mac, verified by deploy-board): identical metrics, G-LAT p99 2.68 ms (forecast_report_ventunoq.{md,json}; git_commit reads unknown because the board copy has no .git). Thresholds were not changed.

Gate ID	Requirement	Split	Measured	Status
G-NI	Campaign decisions/hash unchanged with observer	pytest non-interference	6 tests green; ab2eb619? (scope audit 2; artifacts/scope_audit_2/GOLDEN_DIFF.md)	PASS

G-OFF	Offline runtime, zero network	pytest -k offline	2 tests green	PASS
G-LAT	p99 full observer tick <= 50 ms	test	p99 = 2.68 ms on Ventuno Q (0.91 ms Mac), <= 8 evals/tick (<= 16 in OOD mode)	PASS
G-ACC-RAMP	ratio MAE@30 <= 0.85	test	0.785	PASS
G-ACC-STEADY	ratio MAE@30 <= 1.05	test	0.858	PASS
G-COV	90% band coverage in [0.85, 0.95]	test	0.908	PASS
G-RECALL	recall >= 0.90 (WARN active at crossing)	test	1.00	PASS
G-LEAD	median >= 20 s, p10 >= 10 s	test	median 60.5 s, p10 37.9 s	PASS
G-FA	<= 1.0 false WARN / h	test + control 1 h	0.115 / h (2 in 17.33 h)	PASS
G-OOD	recall >= 0.75 and FA <= 3 / h	ood	recall 0.90, FA 1.5 / h (was 9.7 before the OOD guard)	PASS
G-ENV	false-open windows = 0 with guard	7-day backtest, 169 h	0 (guarded) vs 20 (raw)	PASS

OOD guard (forecast/ood.py). When the live one-step residual scale exceeds 1.9x its calibration median (exit < 1.6x), the guest enters OOD mode: (1) widened-band-only risk stays at WATCH; WARN needs the point forecast + 0.5x the calibrated band to cross the physical limit within 45 s; (2) the displayed band is widened by the live/calibrated residual ratio; (3) that risk must persist 6 s (not 3 s). The sentence states the mode. Thresholds unchanged; the 0.5 / 6 s settings were chosen on a held-out OOD validation set (seeds 6000-6119; 0.7 FA/h, recall 0.90), not on the benchmark ood split. In-distribution behaviour (Golden Gate replay, test split) is unchanged except one fewer false WARN on spikes.

4. Golden Gate Replay (what the operator console shows)

GET /api/forecast is a cached REPLAY of scenario_B_golden_gate_recorded.yaml with SIMULATED telemetry (labelled in the payload and UI), not a live feed. Actual alerts:

> "G5 Emergency Comms & Edge Compute Node rising +8 W/s; upper 90% bound reaches 49.5 kW (110% of declared 45.0 kW) in ~39 s; point forecast stays below within 60 s (90% band 39->60 s); binding: power G5 feeder 49.5/49.5 kW. Advisory only -- governor action unchanged." (T = 260 s)

The reactive GUEST_OVERCONSUMPTION fires at T = 297 s exactly as without the observer, so the real advance notice is ~37 s, driven by the upper band (the point forecast alone would not have warned). A second WARN at T = 1540 s with a flat slope is a likely false alarm and is reported as such.

Guarded windows from the base state show no G4 window in the 48 h outlook: VEHICLE_REPLACEMENT_AT_SEA is closed by weather (guarded sea state 4 > limit 3 every hour); LAUNCH_RECOVERY is closed in all 49 hours by operational blockers (G3 acoustic survey quiet, heading not into wind), plus wind > 25 kn in 18 hours. The API/UI print these reasons instead of a fabricated window.

5. Explicit Limitations & Mandatory Honest Disclosures

* 1. Recorded Public Data Snapshot & Offline Runtime: Recorded public data snapshot (UTC 2026-10); runtime is offline with zero live network calls.
* 2. Coastal Model Resolution & Empirical Guard Bands: Open-Meteo is a coastal model product at ~8 km resolution; validated against NDBC 46237/46026/FTPC1 over 7 days (169 hourly pairs); guard bands are the derived 90% error quantiles: +7.34 kn wind, +0.24 m Hs, +0.51 deg C SST (guard_bands.json, status DERIVED). Without the guard the backtest gives 20 false-open hours; with it, 0.
* 3. Synthetic Telemetry & Advisory Non-Interference: Consumption telemetry is synthetic (seeded); /api/forecast is a labelled replay. The forecaster is advisory and never changes the governor’s decision.
* 4. Thermal Baseline Adherence: At Golden Gate SST (~16 deg C), cooling is not the binding constraint; 14-day admission is still judged at 30 deg C design worst-case.
* 5. Known trade-off: in OOD mode WARN is more conservative (recall 0.90 vs 1.00 in distribution; 4 of 40 OOD ramps warned < 10 s ahead). G-LAT on the Ventuno Q is 2.68 ms p99, far below 50 ms.